

Honeywell

**TPS Builder
CL Editor
User's Guide**

BL21-400

TPS R400

3/2007

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About This Document

This document describes the Honeywell Control Language (CL) Editor functionality and procedures for using CL Editor.

Release Information

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References

The following list identifies all documents that may be sources of reference for material discussed in this publication.

Document Title	Doc ID
<i>TPS Builder Installation, Implementation, and Administration Guide</i>	BL08
<i>TPS Builder Configuration Tool User's Guide</i>	BL20

Contacts

World Wide Web

The following lists Honeywell's World Wide Web sites that will be of interest to our industrial automation and control customers.

Honeywell Organization	WWW Address (URL)
Corporate	http://www.honeywell.com
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1. CL Editor Introduction

1.1 Overview

Scope

The following sections describe the Honeywell Control Language (CL) Editor functionality and procedures for using CL Editor.

Codewright

Borland Codewright, version 7.5 is the underlying application for CL Editor. It is provided by Honeywell with TPS Builder, and must be installed before you can use CL Editor. This document does not include information about installing or using Codewright. Codewright has its own documentation and online Help. Consult these for information on using the application.

Audience

The information in the following sections is for those experienced with CL.

If you are not experienced in this area, please consider appropriate training. Check the Honeywell Automation College website for applicable training programs:

<http://www.automationcollege.honeywell.com/>.

1.2 CL Editor Capabilities and Features

Capabilities

The Honeywell Control Language (CL) Editor is a state-of-the-art editor that allows you to efficiently write CL programs. Based on Borland's Codewright, CL Editor can be accessed from within TPS Builder, or can be used without TPS Builder.

In addition to creating CL programs, the CL Editor has a built-in checking module, the CL Checker. This checker searches for syntax and semantic errors.

Features

The CL Editor provides the following features:

- Standard Windows editor features
 - File manipulation functions
 - Edit functions
 - Tool bar
- ChromaCoding: Different colors can be set for various elements of the CL language's syntax: comments, keywords (reserved words), strings, and numbers
- Template expansion: Typing certain words or abbreviations will cause a template to be inserted into the CL file
- Online Help for syntax and semantic errors
- TPS Builder name browser support
- Opening include files

CL Checker

The CL Checker provides the following features:

- Complete syntax checking for TPS system Process Manager (PM), Advanced Process Manager (APM), and High-performance Process Manager (HPM); Application Module and Application Module^x (in this manual, “AM” refers to both) nodes; and the Multifunction Controller (MC)
- Semantic checking for PM, APM, HPM, AM, and MC nodes of the TPS system
- Checking against the TPS Builder database
- Enables backbuilding of custom data segments, parameter lists, and enumeration types within TPS Builder

2. Getting Started with CL Editor

2.1 Overview

Section contents

This section explains CL Editor installation, how to start CL Editor, and how to exit the application.

System requirements

CL Editor is installed as part of TPS Builder. Ensure that the system meets all requirements stated in the *Basic Installation Requirements* of the *TPS Builder Installation, Implementation & Administration Guide*.

CL Editor requires Borland Codewright version 7.5 or later. This application is provided with TPS Builder, or you may use your own installed version.

2.2 Installation

Description

CL Editor installation requires installation of both Codewright and TPS Builder (including the CL Editor option). If you do not install CL Editor during the initial TPS Builder installation, you can install it later by following TPS Builder installation instructions and choosing only the CL Editor option.

Procedure

Perform the following steps to install CL Editor:

Step	Action
1	If Codewright 7.5 or later is not installed on your computer, install it using the package provided with TPS Builder. Follow the instructions in the Codewright <i>Getting Started</i> guide.
2	Install TPS Builder as explained in the <i>TPS Builder Installation, Implementation, and Administration Guide</i> , ensuring that you choose CL Editor on the Select Components screen.
3	Complete TPS Builder installation.

CL Editor will be available from within TPS Builder and from the TPS Builder program group.

If you do not install CL Editor when TPS Builder is first installed, it can be added later. Refer to the *"Install CL Editor"* procedure in the "Software Installation" section of the *TPS Builder Installation, Implementation & Administration Guide*.

2.3 Start/Exit CL Editor

Start CL Editor

Use one of the following methods to start the CL Editor:

Method	Description
From within TPS Builder	Refer to the <i>TPS Builder Configuration Tool Guide</i> , Working with Strategies section, <i>Associating/Disassociating a Program with a Strategy</i> .
As a standalone application	Click the Start button, then choose Programs > TPS Builder > CL Editor.

Exit CL Editor

To exit CL Editor, choose File > Exit.

If program changes were made, but not saved, Codewright asks whether you want to save changes. Answer "Yes" to save the changes, or "No" to keep the file as it was previously saved.

3. Using CL Editor

3.1 Overview

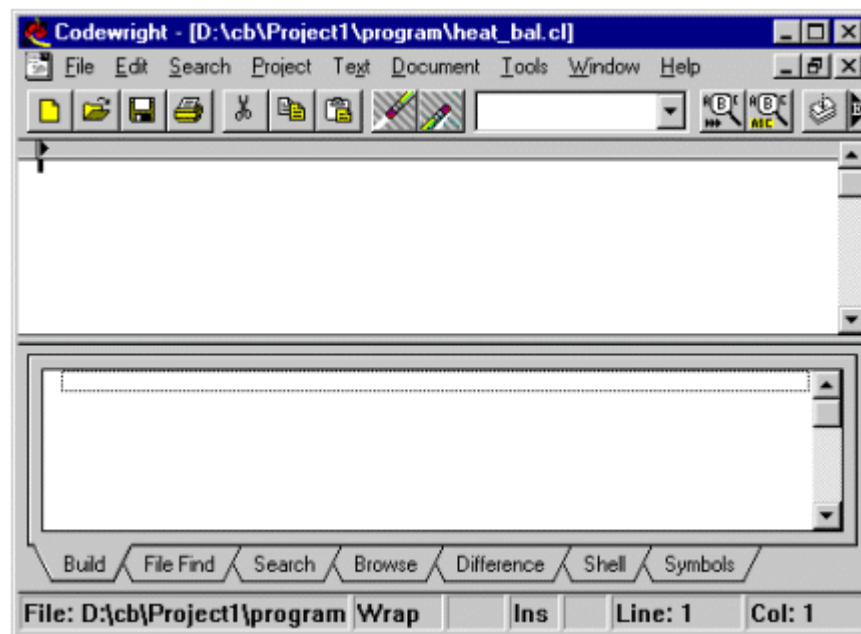
Section contents

The following features of CL Editor are described in this section:

- Basic File Operations
- ChromaCoding
- Template Expansion
- Open Include File
- Checking CL Files
- Online Help
- TPS Strategy Browser
- TPS Reference Manuals
- Word Wrap

CL Editor workspace

CL Editor displays CL programs in the top Text window and the error messages in the bottom Output window. More than one program file can be open in the Text window.



3.2 Basic File Operations

Description

CL Editor is an extension within Codewright, a programmer's editor that offers many features. Refer to the Codewright documentation and online Help for information on using Codewright's features.

You can access CL Editor from within TPS Builder, or separately from TPS Builder. Using CL Editor from within TPS Builder provides two benefits:

- The program you create is associated with the desired TPS Network AM Custom, Switch, and Regulatory; PM-family Process Module (ProcMod); or MC ProcMod data point.
- Program checking includes name checking in accordance with the TPS Builder project that contains the point.

File management

It is recommended that you keep TPS Builder-related CL programs in the *Projectx*\program folder, where *Projectx* is the name of your project. (The complete folder path depends on your TPS Builder installation.)

Save new files to this location, and move existing files there. (You can maintain these program files in other locations if desired.)

Create a new CL program in TPS Builder

Perform the following steps to create a new CL program from within TPS Builder:

Step	Action
1	Start TPS Builder.
2	Open the project or drawing containing the strategy (AM Custom, Switch, or Regulatory, PM-family ProcMod, or MC ProcMod data point) with which you want to associate the program.
3	Right-click the strategy and choose Programs... on the resulting menu.
4	In the Programs dialog, click New.
5	In the resulting dialog, enter a name for the new program. You do not need to include the .cl extension, as TPS Builder will do that.
6	This action starts Codewright with CL Editor extensions, and creates a new program with the name you entered. The program is placed in the <i>Projectx</i>\program folder, where <i>Projectx</i> is the name of your project. (The complete folder path depends on your TPS Builder installation.)
7	Create the CL program, save, and check/correct it using the CL Checker. See for information on the Checker.
8	When you have finished working with the CL program, save it, exit CL Editor, and return to TPS Builder. The program will be included in the strategy's Associated Programs listing.

You must exit CL Editor in order to return to TPS Builder.

Edit an existing CL program in TPS Builder

To edit an existing CL program from within TPS Builder, the program must first be associated with a strategy. To associate an existing program with a TPS Builder strategy, refer to the TPS Builder online Help.

Perform the following steps to edit a CL program from within TPS Builder:

Step	Action
1	Start TPS Builder.
2	Open the project or drawing containing the strategy block (for example, representing a ProcMod point) to which the program to be edited is associated.
3	Right-click the strategy and choose Programs... on the resulting menu.
4	In the Programs dialog, select (highlight) the program(s) to be edited. If more than one program is to be edited, click a program, then Ctrl+click additional programs.
5	Click the Edit button. Codewright with CL Editor extensions will start and the program(s) will open.
6	Edit the CL program(s), save, and check/correct using the CL Checker. For information on using CL Checker, refer to the CL Programming Tools section, CL Checker.
7	When finished, save the program(s), exit CL Editor, and return to TPS Builder.

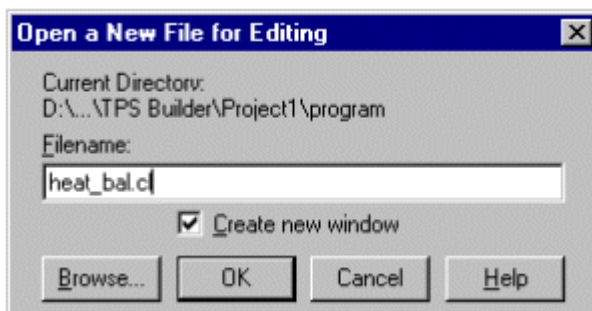
You must exit CL Editor in order to return to TPS Builder.

Create or edit CL programs using the standalone CL Editor

When using CL Editor separately from TPS Builder, the program will not be associated with a TPS Builder strategy (unless this action was done during a previous TPS Builder session). Also, no checking against the TPS Builder database will occur unless you manually enter the related project as explained in this section, CL Programming Tools, Include checking against TPS Builder database.

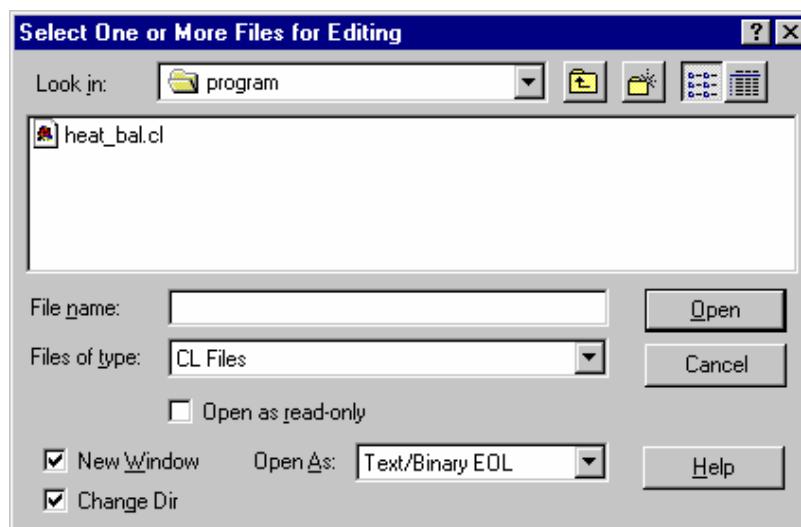
Perform the following steps to use CL Editor separately from TPS Builder:

Step	Action
1	Choose Start > Programs > TPS Builder > CL Editor. This action starts Codewright with CL Editor extensions.
2	Create a new CL program by choosing File > New, or open an existing program by choosing File > Open.
3	For new programs, in the resulting dialog, enter a name for the new program. Include the TPS Builder project folder. You can type the folder, for example, <i>projectname\newfile.cl</i> , where <i>projectname</i> is the TPS Builder project and <i>newfile.cl</i> is the program name. Or, you can click the Browse... button and locate the folder. Click OK to open the new program.



- | | |
|---|--|
| 4 | For existing programs, navigate to the desired program and double-click its name to open it. |
|---|--|
-

Step	Action
------	--------



- | | |
|---|--|
| 5 | Create or edit the program as needed. |
| 6 | Save the program. |
| 7 | If desired, use the CL Checker to check the program. For information on using CL Checker, refer to this section, CL Programming Tools, CL Checker. |
| 8 | Evaluate the error messages in the Output window, and correct program if necessary. Refer to CL Programming Tools in this section, Error correction. |
-

3.3 CL Programming Tools

ChromaCoding

CL Editor gives color to keywords, comments, strings, and literals as you type. By default, the feature is enabled. Perform the following steps to set this feature:

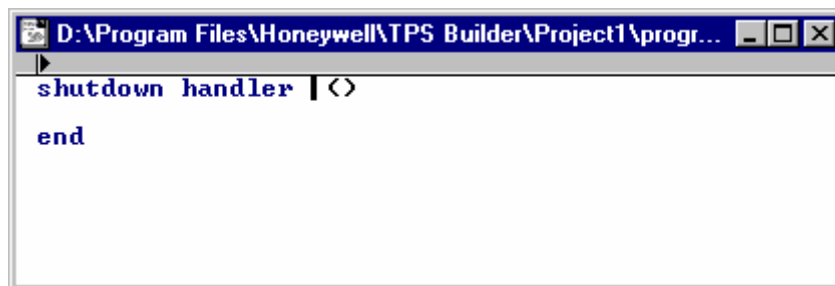
Step	Action
1	Choose Document > Language. The Language dialog box is displayed.
2	Click the Edit tab.
3	Under ChromaCoding group box, check the Language dependent option to enable ChromaCoding.

Template expansion

This feature allows you to type certain words or abbreviations in CL Editor that will cause a CL template to be inserted. For example, typing “IF” followed by a space, causes “IF THEN” to be inserted, and the cursor positioned following the word “IF.”

CL Editor inserts templates for control structures as you type. Perform the following steps to use this feature in a CL program file:

Step	Action
1	Place the cursor in column 1.
2	Type the needed abbreviation as listed in the following table (shortest is fastest).
3	After the template is inserted, complete programming as needed.



Template expansions are listed in the following table. For each item listed

- The first three letters of any word is all that is necessary to invoke the template, but CL Editor will accept 3+ characters up to the entire word. The first three letters are underlined in the following table.
- **A space must be typed after the abbreviation.**
- All basic forms of capitalization are possible (that is, If, IF, and if are all treated the same), even though all “caps” alone are shown here.
- An ampersand (&) in the following table listing, indicates the cursor’s position after the template is added.

When you type this...	This is added to your CL file
BLOCK	BLOCK & () END
<u>CALL</u>	CALL & ()
CUSTOM	CUSTOM &END CUSTOM
EMERGENCY	EMERGENCY HANDLER & () END
HOLD	HOLD HANDLER & () END
<u>IF</u>	IF & THEN
LOOP	LOOP FOR & IN REPEAT
PACKAGE	PACKAGE &END PACKAGE
PARAM_LIST	PARAM_LIST END &
READ	READ & FROM
SEND	SEND &:
SEQUENCE	SEQUENCE & () END
SHUTDOWN	SHUTDOWN HANDLER & () END
<u>SUBROUTINE</u>	SUBROUTINE & () END

When you type this...	This is added to your CL file
WRITE	WRITE & FROM

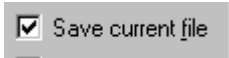
Turn template expansion on or off

By default, the feature is enabled. Perform the following steps to turn this feature on or off:

Step	Action
1	Choose Document > Language. The Language dialog box is displayed.
2	Choose the Edit tab.
3	In the Enhanced Editing group box, choose (checkmark) Template Expansion to turn it on, or clear the checkmark to turn off the feature.
4	Click OK to implement your choice.

CL Checker

Perform the following steps to use CL Checker:

Step	Action
1	In CL Editor, open the CL program to be checked (if it is not already open).
2	If you make changes to the program, make sure you save it before running CL Checker. It is important that you save the file before checking it, as the Checker uses the saved version of the file. One way you can be certain is to set a Codewright option, which causes any modifications contained in the current file to be written to disk before a command is executed. Perform the following steps to set this option: 1) On the Codewright Project menu, click Properties... 1) In the Properties dialog, click the Tools tab. 1) Select the Save current file option:  1) Click OK.
3	Use one of the following methods to run CL Checker: – Click the Check button, on the tool bar

Step	Action
	– Choose Project > Check – Press Ctrl+Shift+F10

Evaluate error messages displayed in the Output window. Refer to Appendix A, Overview of this document for a list of error messages. For information on correcting errors, refer to CL Programming Tools in this section, Error correction.

Include checking against TPS Builder database

If you run CL Editor from within TPS Builder, checking against the TPS Builder database is automatic and includes:

- checking that referenced points exist within the TPS Builder project
- checking that referenced parameters exist for referenced points
- type checking of referenced points

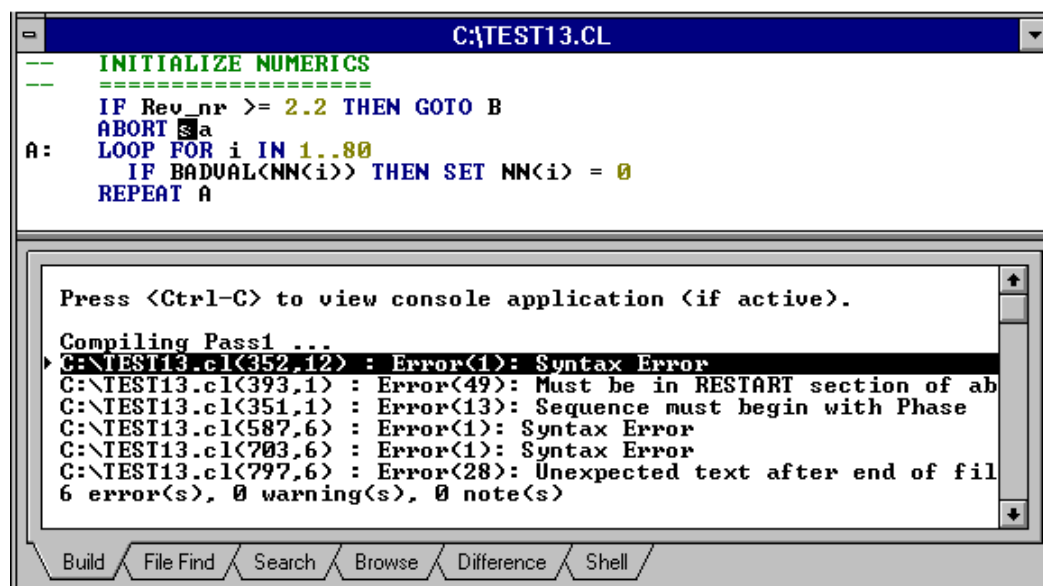
If you use the CL Editor standalone, these checks will not occur unless you perform the following steps:

Step	Action
1	In CL Editor, open the CL program to be checked.
2	From the Project menu, choose Properties....
3	In the Project dialog, click the Tools tab.
4	Make sure Project and Compile are selected, then click the Compiler... button.
5	If the File type is not .cl, click .cl.
6	In the Compile field, make the following entry, where <i>Projectx</i> is the name of your TPS Builder project: <code>C:\Program Files\Honeywell\TPS\Tools\Builder\clcheck.exe %b%e /p "Projectx"</code>
	If you installed TPS Builder in the default location, then substitute that path in the entry above.
7	Click OK in two successive dialogs.
8	Click the Check button.

Error example

This figure shows the Output window after the CL Checker has run, and errors that have been found. Note that error messages include the following information:

- Program location and filename
- Line number and column where the error exists
- Error number and description



Error correction

The CL Editor can help you locate and correct errors quickly. Use any of the following methods to locate the line in the program that contains a specific error:

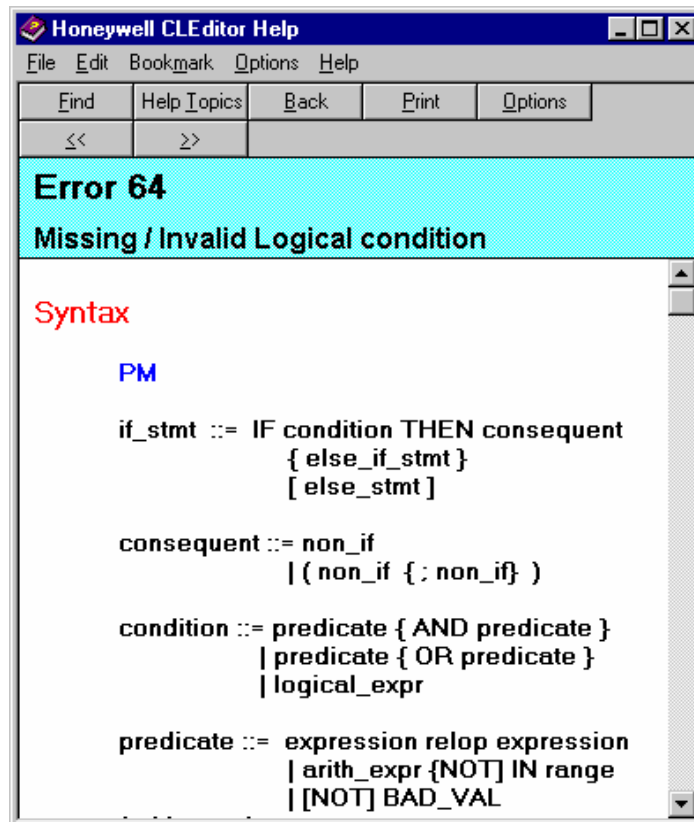
- In the Output window, double-click the error message (which contains an error number) of interest. CL Editor will jump directly to the referenced line in the CL program. The point at which the error occurs is highlighted.
- In the Output window, position the cursor on an error message and press Enter. This action also takes you to the line containing the error.

- Click the *ERR* button in the tool bar. It takes you to the first error in the Output window as well as to the point where the error occurred. Clicking again on the *ERR* button takes you to the next error message.

For information about the error, access CL Editor's online Help, see the next topic.

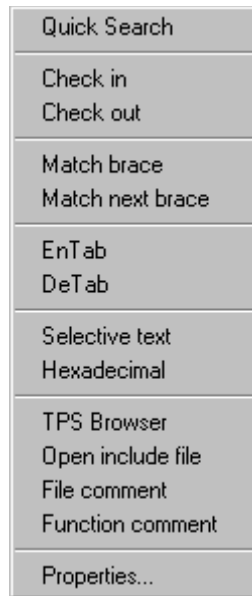
Online Help

CL Editor provides help information for syntax and semantic errors that it displays. For information about a particular error, simply position the cursor on the line containing the error, in the output window, and press F1. Detailed information about the error displays in a separate Help window, as shown in the following example.



Open include file

The following menu appears when your cursor is in a directive in your CL program (do not highlight anything in the directive) and you right-click:



An include file will be opened when the following CL directives are used:

1. `%INCLUDE_SOURCE <filename>`
2. `%INCLUDE_EQUIPMENT_LIST <filename>`

To open an include file, right-click (do not highlight) anywhere on the directive, and select Open include file on the resulting menu.

TPS Strategy Browser

When you need to refer to a TPS name in a CL program but do not know the strategy block or parameter name, you can access the TPS Strategy Browser.

The TPS Strategy Browser can be invoked by right-clicking anywhere in the Text window and selecting TPS Browser from the resulting menu.

The TPS Strategy Browser allows you to browse the TPS Builder project hierarchy and select a TPS name (point or point.parameter). For details on using the TPS Strategy Browser, see the TPS Builder online Help.

When you choose a name within the TPS Strategy Browser and click OK, CL Editor inserts the name at the current position of the cursor in your CL program.

Word wrap

Word wrap causes text to wrap to the next line when it reaches the right margin mark (the 80th column). To enable word wrap, select Word Wrap in the Text menu.

For details about Codewright's word wrap features, see this topic in the Codewright documentation or Help. Settings are made on the Document > Language > Edit tab.

Characters beyond the 80th column cannot be seen in the editor included with the engineering personality. Files with more than 80 characters, however, will compile fine.

4. Appendix A – CL Checker Messages

4.1 Overview

Appendix contents

This appendix contains error messages generated by the CL Checker, including the following:

- PM/APM/HPM Program Error Messages
- AM Program Error Messages
- MC/AMC Program Errors
- Project-Related Messages
- Backbuild Messages

Checking phases

CL Checker has basically two passes:

- Pass 1: A phase of CL Checker activity in which CL Checker checks for syntax errors.
- Pass 2: A phase of CL Checker activity in which CL Checker checks for semantic errors. Pass 2 occurs only if Pass 1 generates no syntax errors.

4.2 PM/APM/HPM Program Error Messages

The following error messages are generated by CL Editor for PM/APM/HPM programs.

Pass 1 errors

1. Syntax Error
2. Routine Out of Order
3. Sequence heading Expected here
4. Sequence Identifier expected here
5. Error in Point Description
6. '(' expected
7. ')' expected
8. End of Line expected
9. POINT expected
10. ';' expected
11. BDP Identifier expected here
12. Phase Identifier expected here
13. Sequence must begin with Phase
14. Must contain executable statements
15. Step Identifier expected here
16. Error in EXTERNAL declaration
17. Error in LOCAL declaration
18. 'AT' expected
19. '=' expected
20. LABEL expected here
21. ':' expected
22. Error in LOOP statement

- 23. LOOP counter expected here
- 24. 'IN' expected here
- 25. LOOP Identifier expected
- 26. Missing END
- 27. Unexpected text after end of file
- 28. ELSE IF cannot begin a statement
- 29. ELSE cannot begin a statement
- 30. Error in IF statement
- 31. Error in Condition
- 32. Error in ELSE IF statement
- 33. Error in Else statement
- 34. Error in SET statement
- 35. Error in SEND statement
- 36. RESUME must have PHASE destination
- 37. Error in READ statement
- 38. Error in GOTO statement
- 39. Error in WAIT statement
- 40. Error in INITIATE statement
- 41. Error in WRITE statement
- 42. Error in REPEAT statement
- 43. Error in CALL statement
- 44. Missing closing quote
- 45. HANDLER Identifier expected here
- 46. HANDLER expected
- 47. Must be in RESTART section of abnormal handler

- 48. SUBROUTINE Identifier expected here
- 49. Source file contains no code
- 50. Error in Subroutine parameters
- 51. Declaration or Phase expected
- 52. Identifier must be PM, APM, or MC - defaulted to PM
- 53. Abnormal handler Identifier expected here
- 54. Error in Headclause
- 55. LOCAL variable expected here
- 56. Error in Type declaration
- 57. Parameter name expected
- 58. Parameter ID must be NN or FL
- 59. Hardware location must be an integer
- 60. Error in Constant assignment
- 61. One dimensional array expected
- 62. External variable expected
- 63. Missing / Invalid Logical condition
- 64. THEN expected here
- 65. PHASE expected here
- 66. '%PAGE' should start at the first column
- 67. Illegal use of '&'
- 68. '%DEBUG' should start at the first column
- 69. '(WHEN ' expected here
- 70. Error in expression
- 71. Error in subroutine parameters
- 72. Statement Error

- 73. Invalid statement
- 74. %INCLUDE_EQUIPMENT_LIST directive only allowed as the first line
- 75. Error in declaration
- 76. Equipment List object file name expected
- 77. Error in range
- 78. Routine must begin with STEP
- 79. Malformed Identifier
- 80. Include Source File expected
- 81. Error Opening File
- 82. RANGE expected
- 83. END Identifier expected
- 84. %INCLUDE_EQUIPMENT_LIST directive is not declared as the first line
- 85. Both KEEPENB and abnormal handlers not allowed in a phase heading
- 86. Phase Heading contains multiple ALARM clauses
- 87. Handler contains multiple RESTARTS
- 88. Cannot Restart in emergency handler
- 89. Use of this file exceeds maximum of 5 nested include source file
- 90. INITIATE.SEQ is not allowed in this sequence program
- 91. INITIATE Point.Param is not allowed in this sequence program

Pass 2 errors

- 1. Identifier must be within 8 characters
- 2. Lower bound must be less than upper bound
- 3. Data Type was not found in system
- 4. Must be BDP Parameter
- 5. Type mismatch between variable type and hardware location

6. Cannot be redeclared
7. Aliasing of Bound data point
8. Undefined symbol
9. Not found
10. Handler already enabled for this condition
11. Unresolved forward reference
12. Loop was never repeated
13. A Loop may have only one REPEAT statement
14. Target does not define a loop
15. Does not match Heading Name
16. Handler condition does not match that defined for the handler
17. Point must be declared in an EXTERNAL statement prior to the LOCAL declaration
18. Labels must be unique within a Program Block
19. Invalid in this context
20. Not a Subroutine
21. Argument multiply defined
22. Too few arguments
23. Too many arguments
24. Type Mismatch
25. Invalid CALL argument
26. Subscript outside bound
27. Wrong data type for this operator
28. This is not an array
29. Cannot modify a local constant or literal
30. Cannot modify an expression

31. This data type may only be compared equal/unequal
32. Wrong number of dimensions
33. Cannot use whole array
34. Logical condition expected
35. Invalid SEND item
36. Too many SEND items
37. String must be within 8 characters
38. Cannot send Arrays
39. Empty SEND statements are not permitted in sequence program
40. Cannot send Conditions
41. Cannot modify IN argument
42. Cannot READ / WRITE this parameter
43. A remote variable is required here
44. Arrays cannot be compared
45. Expression not constant or cannot be evaluated
46. Number cannot be divided by time
47. Cannot multiply time by time
48. Invalid time expression
49. Type number expected
50. Enumeration literal not in this type
51. Enumeration literal expected
52. Invalid Array Index
53. String is valid only in SEND statement
54. Must be an Integer
55. The LOOP counter must be a LOCAL variable
56. Identifier length error
57. String length exceeds maximum characters

4.3 AM Program Error Messages

Pass 1 errors

1. Syntax error
2. '(' expected
3. ')' expected
4. End of line expected
5. ';' expected
6. Must Contain executable statements
7. Error in EXTERNAL declaration
8. Error in LOCAL declaration
9. 'AT' expected
10. '=' expected
11. LABEL expected here
12. ':' expected
13. Error in LOOP statement
14. Loop Counter expected here
15. 'IN' expected
16. LOOP Identifier expected
17. Missing END
18. Unexpected text after end of file
19. ELSE IF cannot begin a statement
20. ELSE cannot begin a statement
21. Error in IF statement
22. Error in Condition
23. Error in ELSE IF statement

- 24. Error in Else statement
- 25. Error in SET statement
- 26. Error in SEND statement
- 27. RESUME must have PHASE destination
- 28. Error in READ statement
- 29. Error in GOTO statement
- 30. Error in WAIT statement
- 31. Error in INITIATE statement
- 32. Error in WRITE statement
- 33. Error in REPEAT statement
- 34. Error in CALL statement
- 35. Missing closing quote
- 36. SUBROUTINE Identifier expected here
- 37. Source file contains no code
- 38. Error in Subroutine parameters
- 39. LOCAL variable expected here
- 40. Error in Type declaration
- 41. Error in Constant assignment
- 42. One dimensional array expected
- 43. External variable expected
- 44. Missing / Invalid Logical condition
- 45. THEN expected here
- 46. <filename> could not be closed
- 47. <filename> could not be deleted
- 48. '%PAGE' should start at the first column

- 49. Illegal use of '&'
- 50. %DEBUG' should start at the first column
- 51. '(WHEN' expected here
- 52. Error in expression
- 53. Error in subroutine parameters
- 54. Statement Error
- 55. %INCLUDE_EQUIPMENT_LIST directive allowed only as the first line
- 56. Error in declaration
- 57. Equipment list object file name expected
- 58. RANGE expected
- 59. Malformed Identifier
- 60. Include source file expected
- 61. <filename> could not opened
- 62. RANGE expected
- 63. END Identifier expected
- 64. Use of this file exceeds maximum of 5 nested include source files
- 65. Heading expected here
- 66. Block Identifier expected here
- 67. Parameter list Identifier expected here
- 68. Invalid %Include_Set filename, must be 6 characters or less
- 69. Quoted string is too long. Maximum length allowed is 78
- 70. String literal expected here
- 71. Only one Custom Data Definition is permitted
- 72. Statements not valid in Custom Data Segments
- 73. Not a legal %RELAX option

- 74. Allowed only before first Block in a Package or in Block data declarations
- 75. Empty custom data segment not permitted
- 76. Include set file expected
- 77. ';' expected
- 78. Declaration or Block expected
- 79. RELAX identifier expected
- 80. Parameter identifier expected
- 81. Invalid attribute
- 82. Numeric literal expected here
- 83. Parameter Declaration expected
- 84. Declaration or statements expected
- 85. One dimensional array expected
- 86. PACKAGE expected
- 87. CUSTOM expected
- 88. Error in Enumeration
- 89. '/' expected
- 90. Point Identifier expected
- 91. Insertion point expected
- 92. Access Key identifier expected
- 93. Function parameters expected
- 94. Error in Function declaration
- 95. Unexpected text after end of line
- 96. Error in CDS headclause
- 97. Access lock identifier expected
- 98. Class identifier expected

- 99. Error in enumeration declaration
- 100. Identifier expected
- 101. POINT/GENERIC expected
- 102. A positive, non - zero integer expected
- 103. Error in Block clause
- 104. Expression expected
- 105. ACCESS expected
- 106. Error in Block declaration
- 107. Error in function definition
- 108. Invalid declaration
- 109. Invalid CL program
- 110. Attribute previously defined
- 111. This array limited to one dimension
- 112. PAUSE statement is not allowed in AM/CL
- 113. RESUME statement is not allowed in AM/CL
- 114. ENB statement is not allowed in AM/CL
- 115. READ statement is not allowed in AM/CL
- 116. WAIT statement is not allowed in AM/CL
- 117. INITIATE statement is not allowed in AM/CL
- 118. WRITE statement is not allowed in AM/CL
- 119. FAIL statement is not allowed in AM/CL
- 120. Parameter declaration not allowed in Subroutines
- 121. External declaration not allowed in Subroutines
- 122. Invalid in Blocks
- 123. Declared with invalid data type

- 124. Illegal use of “%”
- 125. Invalid Access Identifier
- 126. Invalid Class Identifier
- 127. Error in Block heading
- 128. Invalid Access key
- 129. Error in variable declaration
- 130. Subroutine arguments expected
- 131. Source file could be corrupted
- 132. Failed to open registry key
- 133. Failed to read value from the registry
- 134. Error in Enable statement
- 135. '%INCLUDE_SET' should start at the first column
- 136. '%RELAX' should start at the first column
- 137. '%INCLUDE_EQUIPMENT_LIST' should start at the first column
- 138. SEND with WAIT option is invalid in AM / CL
- 139. Error in type definition
- 140. Error in assignment
- 141. Missing / Invalid hardware location
- 142. Function arguments expected
- 143. Assignment expected
- 144. Routine type missing

Pass 2 errors

1. - Attempt to redefine parameter with different type
2. - Attempt to redefine as parameter name
3. %INCLUDE_EQUIPMENT_LIST directive is not supported for CL/AM
4. Defined but not referenced
5. Loop was never repeated
6. Type Enumeration expected
7. Index type mismatch
8. Number of values differs from number of array elements
9. Value does not match parameter type
10. ARRAY expected
11. Cannot modify Point Id - use Data Entity Builder
12. Library Subroutine not permitted in a Package
13. Enumeration literal may not appear here
14. Cannot Send Point Identifiers
15. Point Identifiers cannot be compared
16. Unresolved forward reference
17. Cannot modify function
18. A Loop may have only one REPEAT statement
19. Target does not define a loop
20. Does not match Heading Name
21. Labels must be unique within a Program Block
22. Invalid in this context
23. Argument multiply defined
24. Too few arguments

- 25. Too many arguments
- 26. Type mismatch
- 27. Invalid CALL argument
- 28. Subscript outside bound
- 29. Wrong data type for this operator
- 30. This is not an array
- 31. Cannot modify local constant or enumeration literal
- 32. Cannot modify an expression
- 33. This data type may only be compared equal/unequal
- 34. Wrong number of dimensions
- 35. Cannot use entire array
- 36. Logical condition expected
- 37. Invalid SEND item
- 38. Cannot send Arrays
- 39. Cannot SEND conditions
- 40. Cannot modify IN argument
- 41. Arrays cannot be compared
- 42. Expression is not constant or cannot be evaluated
- 43. Number cannot be divided by time
- 44. Cannot multiply time by time
- 45. Type number expected
- 46. Enumeration literal not in this type
- 47. Enumeration literal expected here
- 48. Array index must be of type number
- 49. Must be an integer

- 50. The LOOP counter must be a LOCAL variable
- 51. <name> not found
- 52. Aliasing of Bound Data point
- 53. Undefined symbol
- 54. <name> not found
- 55. %INCLUDE_EQUIPMENT_LIST directive is not declared as the first line
- 56. Identifier must be within 8 characters
- 57. Lower bound must be less than upper bound
- 58. Data Type was not found in system
- 59. <name> cannot be redeclared
- 60. Data Point Identifier expected
- 61. Loop Index not declared
- 62. Identifier length error
- 63. Conditions are not allowed as CALL arguments
- 64. Cannot SEND Expressions
- 65. Elements of an incoming array argument are not allowed as arguments to another subroutine

4.4 MC/AMC Program Errors

Pass 1 Errors

1. Syntax Error
2. Routine Out of Order
3. Sequence heading Expected here
4. Sequence Identifier expected here
5. Error in Point Description
6. '(' expected
7. ')' expected
8. End of Line expected
9. POINT expected
10. ';' expected
11. BDP Identifier expected here
12. Phase Identifier expected here
13. Sequence must begin with Phase
14. Must contain executable statements
15. Step Identifier expected here
16. Error in EXTERNAL declaration
17. Error in LOCAL declaration
18. 'AT' expected
19. '=' expected
20. LABEL expected here
21. ':' expected
22. Error in LOOP statement
23. LOOP counter expected here

- 24. 'IN' expected here
- 25. LOOP Identifier expected
- 26. Missing END
- 27. Unexpected text after end of file
- 28. ELSE IF cannot begin a statement
- 29. ELSE cannot begin a statement
- 30. Error in IF statement
- 31. Error in Condition
- 32. Error in ELSE IF statement
- 33. Error in Else statement
- 34. Error in SET statement
- 35. Error in SEND statement
- 36. RESUME must have PHASE destination
- 37. Error in READ statement
- 38. Error in GOTO statement
- 39. Error in WAIT statement
- 40. Error in INITIATE statement
- 41. Error in WRITE statement
- 42. Error in REPEAT statement
- 43. Error in CALL statement
- 44. Missing closing quote
- 45. HANDLER Identifier expected here
- 46. HANDLER expected
- 47. Must be in RESTART section of abnormal handler
- 48. SUBROUTINE Identifier expected here

- 49. Source file contains no code
- 50. Error in Subroutine parameters
- 51. Declaration or Phase expected
- 52. Identifier must be PM, APM, or MC - defaulted to MC
- 53. Abnormal handler Identifier expected here
- 54. Error in Headclause
- 55. LOCAL variable expected here
- 56. Error in Type declaration
- 57. Parameter name expected
- 58. Parameter ID must be NN or FL
- 59. Hardware location must be an integer
- 60. Error in Constant assignment
- 61. One dimensional array expected
- 62. External variable expected
- 63. Missing / Invalid Logical condition
- 64. THEN expected here
- 65. PHASE expected here
- 66. '%PAGE' should start at the first column
- 67. Illegal use of '&'
- 68. '%DEBUG' should start at the first column
- 69. '(WHEN ' expected here
- 70. Error in expression
- 71. Error in subroutine parameters
- 72. Statement Error
- 73. Invalid statement

- 74. %INCLUDE_EQUIPMENT_LIST directive only allowed as the first line
- 75. Error in declaration
- 76. Equipment List object file name expected
- 77. Error in range
- 78. Routine must begin with STEP
- 79. Malformed Identifier
- 80. Include Source File expected
- 81. Error Opening File
- 82. RANGE expected
- 83. END Identifier expected
- 84. %INCLUDE_EQUIPMENT_LIST directive is not declared as the first line
- 85. Both KEEPERB and abnormal handlers not allowed in a phase heading
- 86. Phase Heading contains multiple ALARM clauses
- 87. Handler contains multiple RESTARTS
- 88. Cannot Restart in emergency handler
- 89. Use of this file exceeds maximum of 5 nested include source file
- 90. !Box or !DI is not valid in MC
- 91. Point ID cannot be referenced in a local variable in case of MC/AMC
- 92. Badval and Finite are not allowed in MC/AMC
- 93. CALL statement cannot occur in HANDLER or SUBROUTINE in MC/AMC
- 94. Loop control is available only in Step or Subroutine in case of MC/AMC
- 95. Sequence number expected

Pass 2 Errors

1. Identifier must be within 8 characters
2. Lower bound must be less than upper bound
3. Data Type was not found in system
4. Must be BDP Parameter
5. Type mismatch between variable type and hardware location
6. cannot be redeclared
7. Aliasing of Bound data point
8. Undefined symbol
9. not found
10. Handler already enabled for this condition
11. Unresolved forward reference
12. Loop was never repeated
13. A Loop may have only one REPEAT statement
14. Target does not define a loop
15. Does not match Heading Name
16. Handler condition does not match that defined for the handler
17. Point must be declared in an EXTERNAL statement prior to the LOCAL declaration
18. Labels must be unique within a Program Block
19. Invalid in this context
20. Not a Subroutine
21. Argument multiply defined
22. Too few arguments
23. Too many arguments
24. Type Mismatch

- 25. Invalid CALL argument
- 26. Subscript outside bound
- 27. Wrong data type for this operator
- 28. This is not an array
- 29. Cannot modify a local constant or literal
- 30. Cannot modify an expression
- 31. This data type may only be compared equal/unequal
- 32. Wrong number of dimensions
- 33. Cannot use whole array
- 34. Logical condition expected
- 35. Invalid SEND item
- 36. Too many SEND items
- 37. String must be within 8 characters
- 38. Cannot send Arrays
- 39. Empty SEND statements are not permitted in sequence program
- 40. Cannot send Conditions
- 41. Cannot modify IN argument
- 42. Cannot READ / WRITE this parameter
- 43. A remote variable is required here
- 44. Arrays cannot be compared
- 45. Expression not constant or cannot be evaluated
- 46. Number cannot be divided by time
- 47. Cannot multiply time by time
- 48. Invalid time expression
- 49. Type number expected

- 50. Enumeration literal not in this type
- 51. Enumeration literal expected
- 52. Invalid Array Index
- 53. String is valid only in SEND statement
- 54. Must be an Integer
- 55. The LOOP counter must be a LOCAL variable
- 56. Identifier length error
- 57. String length exceeds maximum characters
- 58. Data Point must be a MC point
- 59. Data Point must be on the same Hiway as Bound Data Point
- 60. Remote variables in READ/WRITE/INITIATE must be on the same C-LINK as Bound Data Point

4.5 Project-Related Messages

The following messages are generated when the CL Editor is used from within TPS Builder, or the project is defined for the CL Editor as explained in the Using the CL Editor section, CL Programming Tools, Include checking against TPS Builder database

Pass 2 errors

1. Data point does not exist
2. Not a Process Module Point
3. Must be an APM resident point
4. Must be a PM resident point
5. Point must be on the same UCN as the Bound Data point
6. Must be an AM resident point
7. Illegal Compiler Directive
8. Valid Options - clcheck /I FileName /P ProjectName /S StrategyName
9. Project Name is not given. No database checking will be done
10. Project does not exist in the database. Validation against database will not be done
11. Could not log on to the control session. Validation against database will not be done
12. Exceeded maximum number of arguments
13. File name expected
14. Parameter OP not on this data point
15. Parameter OP should be of discrete type
16. Cannot store to an IOL parameter in a READ statement
17. No corresponding item for this variable
18. Too many variables in this statement
19. Must be a HPM resident point
20. Value not set for array parameter <name>
21. <name> already exists in the database, not overwritten

- 22. Parameter List <name> was not found in the database.
- 23. Must be a MC resident point

4.6 Backbuild Messages

- 1. Value not set for array parameter <param name>
- 2. <name> already there in the database, it is not overwritten
- 3. Backbuilding <name>
- 4. Backbuild started!
- 5. Backbuild Ended!
- 6. User cancelled CDS Backbuild

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